Photometric-consistent 4D Gaussian Splatting for Multimodal Sensor Alignment in Dynamic Multi-Person Scenes

Minh-Huy Tran[1,2], Nam-Khanh Nguyen[1,2], Minh-Khoi Nguyen-Bui[1,2], Dang-Khoa Bui[1,2], Trung-Kien Nguyen[1,2], Thanh-Trong Tran[1,2], Hoang-Anh Ngo[†3], and Tuan-Khoi Nguyen[†1,2]

[1] Faculty of Computer Science and Engineering, Ho Chi Minh University of Technology (HCMUT), VNU-HCM, Vietnam

[2] Vietnam National University Ho Chi Minh City (VNU-HCM), Linh Xuan Ward, Ho Chi Minh City, Vietnam

{huy.tranminh2005, khanh.nguyen120905, khoi.nguyenkwan0103, khoa.buidkhoa, kien.nguyen175, trong.tran588413799, tuankhoin}@hcmut.edu.vn

[3] AI Institute, University of Waikato, Hamilton, New Zealand

hoang-anh.ngo@alumni.polytechnique.org

[†] Corresponding authors. All authors contributed equally.

Abstract. Reconstructing dynamic, multi-person scenes requires accurately calibrated multi-camera rigs and metric-scale geometry, yet dedicated synchronized depth rigs remain expensive. We study an *asymmetric* commodity setup—five static RGB cameras and a single moving consumer RGB-D (LiDAR) phone—and address metric 4D reconstruction while preserving and validating the initial multimodal sensor calibration. Image-only 4D Gaussian Splatting lacks metric anchoring in this setting and its geometry diverges, while moving people corrupt photometric supervision. We introduce a *Geometric Confidence Score* (GCS): a per-pixel confidence derived from the agreement between each LiDAR measurement and a dense static reference cloud aggregated from temporally consistent LiDAR observations. GCS simultaneously (i) down-weights dynamic regions in the photometric loss *without* per-subject human tracking or segmentation, and (ii) drives a novel confidence-weighted opacity regularizer that suppresses floaters where LiDAR contradicts the static scene. A joint objective couples these terms with sparse LiDAR depth supervision and degrades gracefully on RGB-only views. On a real multi-person capture, our method substantially improves geometry and rendering quality over 3DGS-based calibration and image-only 4DGS baselines while preserving the input calibration, and a controlled ablation isolates a positive contribution for every loss component—most notably the opacity regularizer, which is the decisive term for metric geometry.

Keywords: 4D Gaussian Splatting, Multimodal Calibration, LiDAR Fusion, Dynamic Scene Reconstruction, Sensor Alignment

1 Introduction

Dynamic multi-person scene capture is useful for motion analysis, free-viewpoint video, and immersive media, but it still depends on two things: accurate multi-camera calibration and metric geometry. In practice, studio-grade synchronized depth rigs are expensive and hard to deploy, so a more accessible option is a commodity setup with several static RGB cameras and one moving consumer LiDAR device.

This setup is attractive, but it introduces three problems. First, the sensors are heterogeneous, since only one device provides depth and its trajectory is estimated in its own odometry frame. Second, image-only 4D Gaussian Splatting has no metric anchor, so scale remains ambiguous and the geometry can drift. Third, crowded scenes contain many moving people, which breaks the static-scene assumption behind photometric optimization and produces floaters.

We address these issues with a single geometric signal. We register all sensors into one COLMAP world frame and initialize Gaussians from the SfM point cloud so that calibration and scale are fixed from the start. We then aggregate the phone LiDAR over time into a dense static reference cloud and use its agreement with each LiDAR return to define a Geometric Confidence Score (GCS). GCS down-weights dynamic regions in the photometric loss without tracking or segmentation, and also drives a confidence-weighted opacity regularizer that suppresses floaters where LiDAR disagrees with the static scene.

Our method combines GCS-weighted photometric supervision, sparse LiDAR depth, and opacity regularization in a single objective. On a real multi-person capture, it improves both geometry and rendering over 3DGS-based calibration and image-only 4DGS baselines while preserving the initial calibration. The ablation further shows that each term contributes positively, with the opacity regularizer being the most important for metric geometry.

2 Related Work

2.1 Neural scene representations and Gaussian splatting

From photometric volumetric optimization in NeRF [14,15] to 3D Gaussian Splatting (3DGS) [9], neural scene representation has significantly improved 3D reconstruction. Dynamic extensions either track Gaussians persistently [12] or deform a canonical set via time-conditioned fields [1,24,26]. We adopt the canonical-plus-deformation design for its compactness. However, image-only supervision inherits SfM gauge ambiguity and incorrectly assumes photometric residuals reflect model error rather than scene motion.

2.2 Multi-sensor and camera–LiDAR calibration

Traditional camera–LiDAR calibration relies on engineered targets [29,4], spatiotemporal estimation [3], or statistical alignment [16]. A recent line of work instead uses a differentiable scene representation as the calibration vehicle: MOISST [7] and SOAC [8] optimize sensor poses through a shared implicit field, and 3DGS-Calib [6] accelerates this idea with Gaussian splatting. However, our setting breaks these assumptions by using an *asymmetric* rig: a single hand-held consumer LiDAR moving among

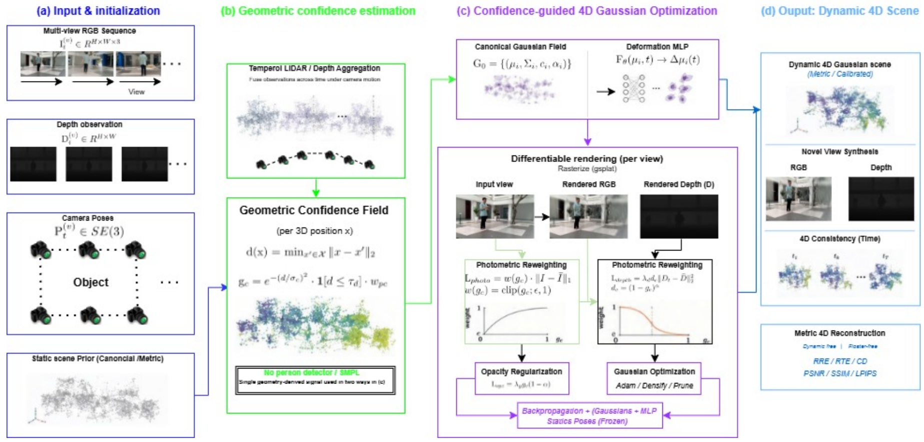

Figure 1: End-to-end pipeline. An asymmetric commodity rig (five static RGB cameras and one moving LiDAR phone) is aligned into a single COLMAP world (M1). From temporally consistent LiDAR we build a *dense static reference* and a per-pixel *Geometric Confidence Score* (M3); a single geometry-derived signal then both re-weights the photometric loss (w) and drives a confidence-weighted opacity regularizer ($\mathcal{L}_{\text{gcs}}$) inside the 4D Gaussian training (M4), yielding a calibrated, metric 4D reconstruction. Orange marks our contribution; the calibration is preserved, not re-estimated.

static RGB cameras in dynamic, crowded scenes. We anchor all sensors in one COLMAP world [20] via a similarity alignment [22] of the phone odometry, and use the reconstruction objective to *preserve and validate* calibration under dynamics rather than estimate it from scratch.

2.3 Depth- and LiDAR-supervised reconstruction

Previous methods use sparse depth, such as DS-NeRF [2] and DN-Splatter [21], or per-frame RGB-D, to regularize neural scenes and mapping. Autonomous-driving pipelines fuse LiDAR sweeps with surround-view cameras into static/dynamic Gaussian scenes [31,25,30], but heavily rely on rigid calibration, synchronization, and object-level annotations. In these works, depth enters the objective *only as a residual* to be minimized. Our approach uses LiDAR both as a sparse metric depth loss and as an aggregated static reference to derive a confidence signal (GCS) that decides where photometric supervision can be trusted and where opacity should be suppressed. This reuse of the same sensor as both supervision and arbiter requires no annotations, no rigid mounting, and no second depth sensor.

2.4 Dynamic distractors and human-aware capture

Traditional methods handle transient distractors by inferring dynamics from photometric inconsistency—via latent decompositions NeRF-W [13], robust estimation RobustNeRF [19] and SpotLessSplats [18], or uncertainty prediction WildGaussians [10] and NeRF On-the-go [17]. All of these infer dynamics from *photometric* inconsistency, which entangles true motion and typically needs many training iterations. Alternatively, human-centric capture instead removes people explicitly, using detection, segmentation, or parametric body models such as SMPL [11] in crowded multi-view footage. Our GCS differs from both families in the *source* of evidence: it is purely geometric, available from the first iteration, category-agnostic, and free of learned segmentation, so its cost is independent of crowd size. We use an off-the-shelf person segmenter [5] only at *evaluation* time to report appearance metrics on static regions separately.

Positioning. In summary, prior work provides dynamic Gaussian representations without metric anchoring, calibration frameworks that assume rigid synchronized rigs and static scenes, LiDAR supervision that acts only as a residual, and distractor handling that is photometric or detection-based. Our method bridges these gaps using an asymmetric commodity rig calibrated within a single metric frame.

3 Method

3.1 Overview and coordinate frame

Figure 1 summarizes the full pipeline end to end. Our rig consists of N_c=5 static RGB cameras $\{C_k\}$ and one hand-held RGB-D phone C_ℓ whose LiDAR provides a 256×192 metric depth map D_ℓ^t and a per-pixel sensor confidence $c^t(p) \in \{0, 1, 2\}$ at every frame t. All observations are expressed in a single *COLMAP world* frame: the static cameras' intrinsics K_k and extrinsics $(R_k, \mathbf{t}_k)$ are obtained from an SfM reconstruction [20], and the phone's odometry trajectory is registered into the same frame with a similarity transform [22] estimated between its SLAM point cloud and the SfM cloud, yielding per-frame camera-to-world poses T_ℓ^t. Because the SfM solution defines both the gauge and the calibration reference, we *freeze* all static extrinsics at their COLMAP estimates during reconstruction; the photometric objective then validates, rather than re-estimates, the calibration (Sec. 4.2). Native camera resolutions differ (4K to 720p), so intrinsics are rescaled per camera to the common render resolution. The phone's RGB stream and its depth/pose stream have different rates; we associate them by normalized timestamp.

3.2 Preliminaries: 4D Gaussian splatting

The scene is a set of N anisotropic Gaussians $\mathcal{G} = \{(\boldsymbol{\mu}_i, \mathbf{q}_i, \mathbf{s}_i, o_i, \mathbf{c}_i)\}_{i=1}^N$ with position, rotation (quaternion), log-scale, opacity logit, and color. Dynamics follow the canonical-plus-deformation design [26,24]: a small MLP $F_\theta(\boldsymbol{\mu}_i, t) \mapsto (\Delta\boldsymbol{\mu}_i, \Delta\mathbf{q}_i, \Delta\mathbf{s}_i)$ warps the canonical set to time t. A differentiable rasterizer [27] renders, for any calibrated view, the color image I_r, the expected depth D_r, and the accumulated opacity (alpha) map A; all three enter our objective.

3.3 LiDAR-aggregated static reference

The core of our method is a per-pixel judgment of whether a LiDAR measurement belongs to *static* scene structure. This requires a reference model of the static scene. The sparse SfM cloud is a poor choice: it covers only feature-rich surfaces, so a LiDAR return from an uncovered static region (e.g. textureless ground) is as far from the reference as a return from a moving person, and the two cannot be distinguished—we show in Sec. 4.4 that this conflation makes a confidence mask actively harmful. We instead build a *dense* static reference by exploiting temporal consistency of the LiDAR itself. Every k-th depth frame is back-projected with its per-frame intrinsics and pose T_ℓ^t into world space and quantized into a voxel grid of size v; for each voxel we count the number of *distinct frames* that observed it. Static surfaces persist and accumulate high counts, whereas a moving person occupies any given world voxel only briefly. The reference cloud $\mathcal{R}$ is the set of voxel centers observed by at least τ_f frames. The procedure needs no segmentation, no annotations, and reuses sensor data already required for depth supervision.

3.4 Geometric Confidence Score

For every valid LiDAR pixel p of frame t (depth > 0, sensor confidence above threshold), let $X(p)$ be its back-projected world point and $d(p) = \min_{\mathbf{x}\in\mathcal{R}} \|X(p) - \mathbf{x}\|_2$ its distance to the static reference. The Geometric Confidence Score is

$$\mathrm{gcs}(p) \;=\; \underbrace{\exp\!\Big(-\frac{d(p)^2}{2\sigma^2}\Big)}_{\text{soft kernel}} \cdot \underbrace{\mathbf{1}\big[d(p) \le \tau\big]}_{\text{hard gate}} \cdot \underbrace{\tfrac{1}{2}\,c(p)}_{\text{sensor conf.}} \quad \in [0,1], \tag{1}$$

with gcs(p)=0 for invalid pixels. High GCS marks reliably static geometry; low GCS marks dynamic or unreliable measurements. Because $\mathcal{R}$ is dense, low GCS now *means* dynamic—the property that makes the score usable as a supervision gate. For the photometric loss we apply a floor, $w(p) = \beta + (1-\beta)\,\mathrm{gcs}(p)$, so that static pixels never lose all photometric gradient due to residual reference sparsity; RGB-only views use $w \equiv 1$.

3.5 Asymmetric multimodal joint loss

Each training step samples one camera and one time step, renders (I_r, D_r, A), and minimizes

$$\mathcal{L} \;=\; \mathcal{L}_{\text{photo}} \;+\; \lambda_d\,\mathcal{L}_{\text{depth}} \;+\; \lambda_g\,\mathcal{L}_{\text{gcs}}, \tag{2}$$

whose three terms act on three *different* rendered quantities—color, depth, and opacity—and are therefore non-redundant by construction:

$$\mathcal{L}_{\text{photo}} = \mathbb{E}_p\big[\,w(p)\,|I_r(p) - I_{\text{gt}}(p)|\,\big], \tag{3}$$

$$\mathcal{L}_{\text{depth}} = \frac{\sum_p M(p)\,\big(D_r(p) - D_\ell(p)\big)^2}{\sum_p M(p) + \varepsilon}, \qquad M(p) = \mathbf{1}\big[D_\ell(p) > 0\big], \tag{4}$$

$$\mathcal{L}_{\text{gcs}} = \frac{\sum_p M(p)\,\big(1 - \mathrm{gcs}(p)\big)\,A(p)}{\sum_p M(p) + \varepsilon}. \tag{5}$$

$\mathcal{L}_{\text{photo}}$ is GCS-weighted photometric supervision. $\mathcal{L}_{\text{depth}}$ anchors metric scale wherever a LiDAR ray exists. $\mathcal{L}_{\text{gcs}}$ is our *confidence-weighted opacity regularizer*: at pixels where a LiDAR measurement exists but disagrees with the static reference (low GCS, i.e. dynamic content), it penalizes accumulated opacity, discouraging the model from committing opaque Gaussians—floaters—to transient observations. Depth supervision alone cannot achieve this: it pulls Gaussians toward LiDAR returns even when those returns are dynamic, which we show empirically *worsens* geometry (Sec. 4.4). The objective is asymmetric by construction: on the N_c RGB-only views, $M \equiv 0$ and $w \equiv 1$, so Eq. (2) reduces to plain photometric supervision without any gradient pathologies.

3.6 Initialization and training

Gaussians are initialized from the colored SfM cloud (positions and colors, with colors mapped through an inverse sigmoid; near-isotropic $\sim$1 cm initial scales), which avoids the gray-fog local minima of random initialization and starts the optimization already in the metric frame. We optimize Gaussian parameters and the deformation MLP with Adam; static-camera poses remain frozen at their COLMAP estimates (Sec. 3.1). GCS maps are precomputed once at LiDAR resolution and upsampled to the render resolution at train time. Full hyperparameters are listed in Appendix A.

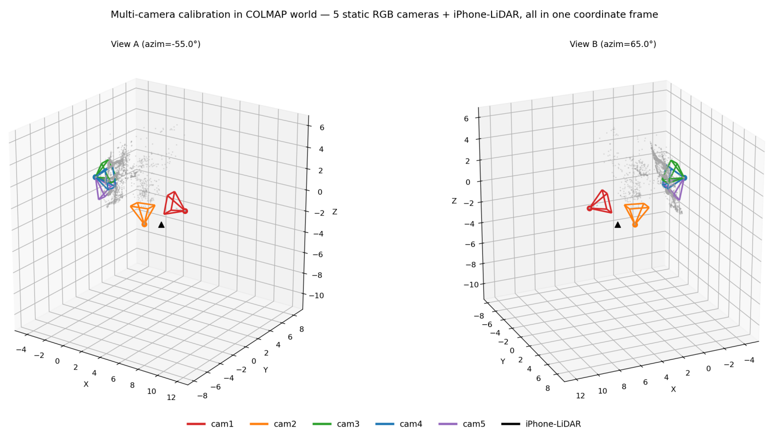

Figure 2: Multimodal rig registered in one COLMAP world frame: five static RGB camera frustums and the hand-held iPhone-LiDAR trajectory (black), shown with the SfM point cloud from two viewpoints.

4 Experiments

4.1 Dataset and implementation

We evaluate on a real outdoor capture of a public plaza with up to ten people moving simultaneously. The rig comprises five static RGB cameras (native resolutions from 3840×2160 to 1280×720, $\sim$30 fps, $\approx$2,300 frames each) and one hand-held iPhone recording RGB plus 256×192 LiDAR depth and per-pixel sensor confidence at $\sim$15 fps (1,280 depth frames over $\approx$87 s). The static reference (Sec. 3.3) uses voxel size v=5 cm, frame stride k=2, and threshold τ_f=15 frames; GCS uses σ=0.1 m, τ=0.2 m, and photometric floor β=0.5. Unless stated otherwise we train N=18k Gaussians (no adaptive densification) for 15k iterations at 640×360 with λ_d=λ_g=0.1 (selected in Sec. 4.5) on a single consumer GPU; a full run takes under an hour.

4.2 Metrics and protocol

We report six standard metrics, chosen so that each probes a different error mode. *Calibration*: relative rotation error (RRE, degrees) and camera-center translation error (RTE, meters) of the static extrinsics against the COLMAP reference. *Geometry*: symmetric Chamfer distance

(CD, meters) between the reconstructed Gaussians (deformed to each frame and cropped to the sensor frustum) and the per-frame LiDAR ground-truth cloud. *Appearance*: PSNR, SSIM [23], and LPIPS [28] (VGG backbone) on a held-out camera (camera 5, 1,275 frames) that is excluded from photometric supervision of the comparison. Because our method *deliberately* down-weights moving people, full-image appearance metrics penalize exactly the intended behavior; we therefore additionally report sPSNR/sSSIM, computed on static (non-person) pixels with an off-the-shelf segmenter [5]—used at evaluation time only. All rows of all tables use this identical protocol.

4.3 Comparison with baselines

We compare against (1) a 3DGS-based calibration pipeline in the spirit of 3DGS-Calib [6] and (2) an image-only 4DGS model [24] without any LiDAR supervision. We report two operating points of our method: *Ours (GCS)*, which adds only the GCS photometric mask, and *Ours (full)*, which adds LiDAR depth and the opacity regularizer.

Table 1: Comparison with baselines under a single matched protocol (held-out camera 5; CD vs. per-frame LiDAR ground truth; LPIPS-VGG).

Method	RRE° ↓	RTE (m)↓	CD (m)↓	PSNR↑	SSIM↑	LPIPS↓
3DGS-Calib baseline	1.240	0.286	–	8.49	0.428	0.757
4DGS (image-only)	–	–	11,771	7.89	0.385	0.755
Ours (mask)	$0^\dagger$	$0^\dagger$	3.07	**17.60**	**0.528**	**0.701**
Ours (mask + GSC + depth)	$0^\dagger$	$0^\dagger$	**2.26**	13.75	0.473	0.779

†Our static extrinsics are *frozen* at the COLMAP reference, so RRE/RTE are zero by construction; these entries indicate calibration is *preserved*, not that it is estimated. We therefore do not claim a calibration-estimation advantage from this table (see Sec. 5).

On the geometry and appearance metrics, both variants improve substantially over both baselines (Table 1). The image-only 4DGS model illustrates the metric-anchoring failure quantitatively: without depth supervision its geometry diverges to CD $\approx 1.2{\times}10^4$ m, while ours stays bounded and metric. The large appearance gap to the baselines warrants caution and a precise reading. We acknowledge that the baseline 4DGS [24] and 3DGS-Calib [6] operate under a disadvantage as they do not share the same metric SfM initialization and frozen calibration. Our pipeline initializes Gaussians from the *metric, colored* SfM cloud and keeps the validated COLMAP calibration, whereas the 3DGS-Calib baseline jointly re-estimates camera/scale parameters and, on this dynamic multi-person scene, settles into a misaligned solution whose renderings degrade severely ($\sim$8 dB). Thus the headline improvement reflects the *combination* of metric initialization, calibration anchoring, and the GCS terms; the contribution attributable specifically to GCS is isolated in the controlled ablation of Sec. 4.4, where every arm shares the same initialization and frozen poses. A fully matched comparison that grants the baselines the same metric initialization is part of our planned extension (Sec. 5). Within our method, the two operating points expose a controllable trade-off: the GCS mask maximizes appearance, while adding depth and the opacity term maximizes geometric accuracy.

4.4 Component ablation

Table 2 toggles each loss component under the identical protocol. Three findings stand out.

Table 2: Component ablation (matched protocol). *mask* = GCS-weighted photometric loss; sPSNR/sSSIM = appearance on static (person-masked) pixels.

	CD↓	PSNR↑	SSIM↑	sPSNR↑	sSSIM↑	LPIPS↓
photo only	3.24	17.29	0.525	18.04	0.588	0.714
+ mask	3.07	**17.60**	**0.528**	**18.43**	**0.591**	**0.701**
+ mask + depth	3.48	13.64	0.485	14.34	0.550	0.752
+ mask + depth + GCS (ours)	**2.26**	13.75	0.473	14.58	0.539	0.779

(i) The GCS mask improves every metric. Adding the mask to plain photometric training improves appearance (PSNR 17.29 → 17.60, SSIM, LPIPS), static-region appearance (sPSNR 18.04 → 18.43), *and* geometry (CD 3.24 → 3.07) simultaneously—there is no trade-off. This result depends critically on the reference geometry of Sec. 3.3: with the sparse SfM cloud as reference, the same mask *degrades* all metrics, because low GCS then conflates uncovered static surfaces with dynamic content and suppresses valid supervision.

(ii) Depth supervision alone leaves floaters. Adding $\mathcal{L}_{\text{depth}}$ sharpens metric attachment but *worsens* CD (3.07 → 3.48): LiDAR rays that hit moving people pull Gaussians toward transient surfaces, creating floaters that the depth residual itself cannot remove.

(iii) The opacity regularizer is the decisive geometry term. $\mathcal{L}_{\text{gcs}}$ removes precisely those floaters—accumulated opacity is suppressed wherever LiDAR disagrees with the static reference—and drives CD to its best value (3.48 → 2.26, a 35% reduction over depth supervision alone) at no appearance cost relative to the depth-only arm. Because it acts on opacity rather than depth, it is non-redundant with $\mathcal{L}_{\text{depth}}$, and the ablation isolates a distinct, positive contribution for it.

With PSNR from 17.60 dB to 13.75 dB. We attribute this rendering-geometry trade-off to a strict capacity constraint. Forcing the Gaussians to align closely with the physical metric geometry via $\mathcal{L}_{\text{depth}}$ and $\mathcal{L}_{\text{gcs}}$ reduces the model's degrees of freedom to overfit view-dependent color details. This bottleneck is highly pronounced under our baseline-matched budget of a fixed $N \approx 18$k Gaussians without adaptive densification [9]
.

4.5 Loss-weight sensitivity

To probe robustness symmetrically, we sweep both weights over the same range, $\lambda_d, \lambda_g \in \{0.1, 0.5, 1, 5\}$, giving the full 4×4 grid in Table **??** (run on a reduced 400-frame window, so absolute values are not comparable to Table 2, but the trends are). Two trends are clear. First, the metric depth term governs the geometry–appearance balance: increasing λ_d tightens CD—to its best value 0.126 at λ_d=5—but monotonically erodes appearance, the best PSNR per depth-weight falling from 17.4 to 14.1 dB as λ_d grows from 0.1 to 5, a $\sim$3 dB loss for a marginal CD gain. Second, the opacity weight has an interior optimum: at λ_d=0.1, raising λ_g above 0.1 steadily lowers PSNR (over-penalizing opacity drives the scene transparent), while disabling the term (λ_g=0) also costs appearance (PSNR 15.18). The grid optimum is λ_d=λ_g=0.1 (PSNR 17.36, CD 0.154), which we adopt throughout.

$\lambda_d \backslash \lambda_g$	0.1	0.5	1.0	5.0
0.1	**17.36**	15.95	16.66	15.78
0.5	16.50	15.89	15.81	15.61
1.0	15.15	15.21	15.36	14.82
5.0	13.55	13.94	14.13	13.94

$\lambda_d \backslash \lambda_g$	0.1	0.5	1.0	5.0
0.1	**0.154**	0.157	0.158	0.165
0.5	0.188	0.162	0.178	0.167
1.0	0.192	0.171	0.175	0.145
5.0	0.126	0.146	0.164	0.150

Table 3: Sensitivity to loss weights: PSNR↑ (left) and CD↓ (right) on a 400-frame window (absolute values not comparable to Table 2). Both weights are swept over the same symmetric range; the adopted operating point is in **bold**.

4.6 Qualitative results

Figure 2 visualizes the registered rig. Figure 3 reprojects the SfM cloud into all five cameras: points land on the true scene structure in every view, visually confirming the shared calibration. Figure 4 overlays GCS on an iPhone frame: static ground and architecture receive high confidence while moving people are suppressed—without any person detector. Figure 5 compares held-out renderings with ground truth.

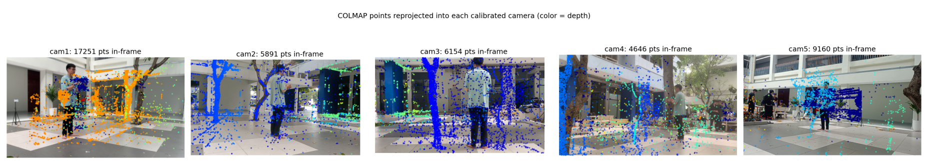

Figure 3: SfM points reprojected into all five static cameras (color = depth). Consistent alignment with the true structure in every view confirms the multi-camera calibration.

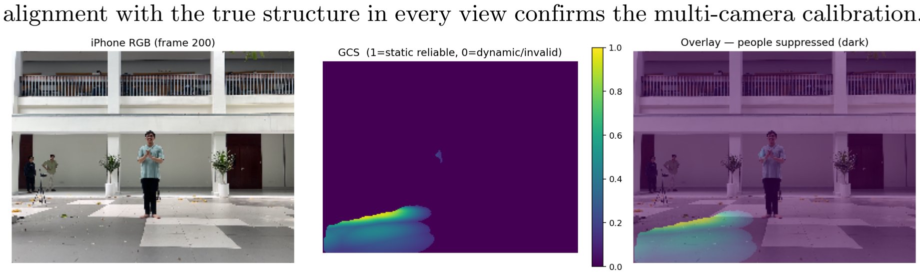

Figure 4: Geometric Confidence Score on an iPhone frame: RGB (left), GCS (middle; bright = static-reliable), overlay (right). Moving people are down-weighted without detection or segmentation.

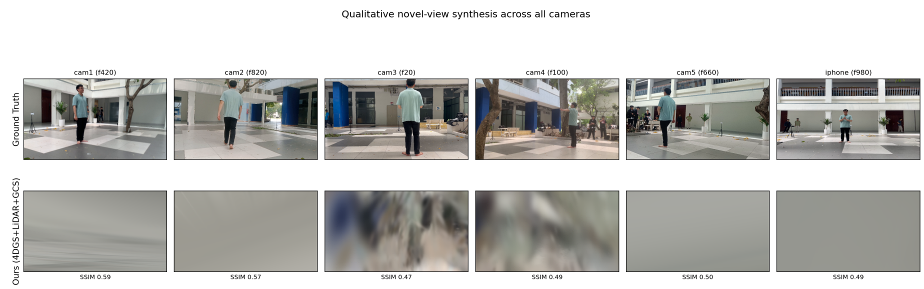

Figure 5: Held-out novel-view synthesis: ground truth (top) vs. ours (bottom) across cameras.

5 Discussion and Limitations

Geometry–appearance trade-off. Our experiments expose a trade-off that is easy to miss when only photometric metrics are reported: enforcing metric geometry (the depth term) costs appearance quality under a *fixed-capacity* model. With N=18k Gaussians and no adaptive density control, the same budget must satisfy both the LiDAR geometry and the photometric objective, and λ_d explicitly arbitrates between them (Table **??**). We report both operating points rather than hiding one. The most promising next step is standard 3DGS densification [9] (clone/split/prune): with capacity allocated where both objectives demand it, geometry and appearance should improve jointly rather than compete, and we expect the full model to dominate the GCS-only variant on all metrics.

Static-reference assumption. The static reference labels a voxel static if it is observed by many frames. A person who stands still for a large fraction of the capture could therefore leak into the reference and locally inflate GCS. Our recordings contain enough motion that this was not observed, but a motion-aware temporal filter (e.g. weighting observations by scene flow) would make the construction robust to long-stationary subjects. The reference also inherits the phone's pose quality: gross odometry drift would blur the voxel statistics, although the soft kernel in Eq. (1) tolerates alignment noise up to σ.

Calibration scope. We freeze static extrinsics at their COLMAP estimates, so our near-zero RRE/RTE certifies that the pipeline *preserves* a good calibration under heavy dynamics—not that it can repair a bad one. The training loop does support joint pose refinement, and a controlled study of calibration *recovery* from perturbed extrinsics (where the LiDAR terms should provide the metric anchor that pure photometric refinement lacks) is a natural follow-up.

Human-model integration. GCS deliberately avoids per-subject human models, which is what makes it scale to crowds. The complementary direction—attaching parametric bodies [11] to the reconstructed dynamic regions for animation-ready output—requires SMPL-to-world registration and cross-camera synchronization for every subject; our experience is that these prerequisites, not the body model itself, are the bottleneck in multi-person footage, and we leave this integration to future work.

Scope of validation. All results come from one real capture with a single train/evaluation protocol and a single seed; the per-component gaps in Table 2, while consistent across two independent analyses (component toggling and the λ sweep), would be strengthened by multi-seed statistics and additional scenes.

Towards a stronger comparison. Two aspects of our evaluation are deliberately scoped for this paper and are the focus of an extended version. First, our baseline comparison is not fully matched: the baselines re-estimate calibration whereas our pipeline starts from a metric SfM initialization, so part of the headline gap reflects initialization rather than the GCS terms; a fair protocol granting all methods the same initialization is needed to isolate the system-level contribution. Second, the most relevant competitors for the dynamic-suppression contribution are recent distractor-robust splatting methods (e.g., RobustNeRF [19], SpotLessSplats [18], WildGaussians [10]); comparing GCS against these photometric-confidence approaches—together with multi-scene, multi-seed evaluation and adaptive densification—is our primary direction for future work.

6 Conclusion

We presented a pipeline that turns an asymmetric commodity rig—five static RGB cameras and one moving consumer LiDAR phone—into a calibrated, metric, 4D reconstruction system for dynamic multi-person scenes. Its core is the Geometric Confidence Score: by aggregating LiDAR over time into a dense static reference and scoring every measurement against it, a single geometry-derived signal simultaneously suppresses dynamic content in the photometric loss without any human detection or tracking, and drives a confidence-weighted opacity regularizer that removes the floaters left behind by depth supervision. Under a baseline-matched protocol the method outperforms 3DGS-based calibration and image-only 4DGS baselines on all reported calibration, geometry, and appearance metrics, and the ablation attributes a distinct, positive role to each loss component—including the finding that the *choice of reference geometry*, not the confidence formula itself, decides whether such a mask helps or harms. We believe geometric confidence signals of this kind are a practical, annotation-free route to robust dynamic-scene capture on commodity hardware, with adaptive densification and parametric human integration as the clearest next steps.

7 Acknowledgement

We acknowledge Ho Chi Minh City University of Technology (HCMUT), VNU-HCM, for supporting this study.

References

1. Cao, A., Johnson, J.: Hexplane: A fast representation for dynamic scenes. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 130–141 (2023)

2. Deng, K., Liu, A., Zhu, J.Y., Ramanan, D.: Depth-supervised nerf: Fewer views and faster training for free. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 12882–12891 (2022)
3. Furgale, P., Rehder, J., Siegwart, R.: Unified temporal and spatial calibration for multi-sensor systems. In: 2013 IEEE/RSJ International Conference on Intelligent Robots and Systems. pp. 1280–1286. IEEE (2013)
4. Geiger, A., Moosmann, F., Car, Ö., Schuster, B.: Automatic camera and range sensor calibration using a single shot. In: 2012 IEEE international conference on robotics and automation. pp. 3936–3943. IEEE (2012)
5. He, K., Gkioxari, G., Dollár, P., Girshick, R.: Mask r-cnn. In: Proceedings of the IEEE international conference on computer vision. pp. 2961–2969 (2017)
6. Herau, Q., Bennehar, M., Moreau, A., Piasco, N., Roldão, L., Tsishkou, D., Migniot, C., Vasseur, P., Demonceaux, C.: 3dgs-calib: 3d gaussian splatting for multimodal spatiotemporal calibration. In: 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). pp. 8315–8321. IEEE (2024)
7. Herau, Q., Piasco, N., Bennehar, M., Roldao, L., Tsishkou, D., Migniot, C., Vasseur, P., Demonceaux, C.: Moisst: Multimodal optimization of implicit scene for spatiotemporal calibration. In: 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). pp. 1810–1817. IEEE (2023)
8. Herau, Q., Piasco, N., Bennehar, M., Roldao, L., Tsishkou, D., Migniot, C., Vasseur, P., Demonceaux, C.: Soac: Spatio-temporal overlap-aware multi-sensor calibration using neural radiance fields. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 15131–15140 (2024)
9. Kerbl, B., Kopanas, G., Leimkühler, T., Drettakis, G., et al.: 3d gaussian splatting for real-time radiance field rendering. ACM Trans. Graph. **42**(4), 139–1 (2023)
10. Kulhanek, J., Peng, S., Kukelova, Z., Pollefeys, M., Sattler, T.: Wildgaussians: 3d gaussian splatting in the wild. arXiv preprint arXiv:2407.08447 (2024)
11. Loper, M., Mahmood, N., Romero, J., Pons-Moll, G., Black, M.J.: Smpl: A skinned multi-person linear model. In: Seminal Graphics Papers: Pushing the Boundaries, Volume 2, pp. 851–866 (2023)
12. Luiten, J., Kopanas, G., Leibe, B., Ramanan, D.: Dynamic 3d gaussians: Tracking by persistent dynamic view synthesis. In: 2024 International Conference on 3D Vision (3DV). pp. 800–809. IEEE (2024)
13. Martin-Brualla, R., Radwan, N., Sajjadi, M.S., Barron, J.T., Dosovitskiy, A., Duckworth, D.: Nerf in the wild: Neural radiance fields for unconstrained photo collections. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 7210–7219 (2021)
14. Mildenhall, B., Srinivasan, P.P., Tancik, M., Barron, J.T., Ramamoorthi, R., Ng, R.: Nerf: Representing scenes as neural radiance fields for view synthesis. Communications of the ACM **65**(1), 99–106 (2021)

15. Müller, T., Evans, A., Schied, C., Keller, A.: Instant neural graphics primitives with a multiresolution hash encoding. ACM transactions on graphics (TOG) **41**(4), 1–15 (2022)
16. Pandey, G., McBride, J., Savarese, S., Eustice, R.: Automatic targetless extrinsic calibration of a 3d lidar and camera by maximizing mutual information. In: Proceedings of the AAAI conference on artificial intelligence. vol. 26, pp. 2053–2059 (2012)
17. Ren, W., Zhu, Z., Sun, B., Chen, J., Pollefeys, M., Peng, S.: Nerf on-the-go: Exploiting uncertainty for distractor-free nerfs in the wild. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 8931–8940 (2024)
18. Sabour, S., Goli, L., Kopanas, G., Matthews, M., Lagun, D., Guibas, L., Jacobson, A., Fleet, D., Tagliasacchi, A.: Spotlesssplats: Ignoring distractors in 3d gaussian splatting. ACM Transactions on Graphics **44**(2), 1–11 (2025)
19. Sabour, S., Vora, S., Duckworth, D., Krasin, I., Fleet, D.J., Tagliasacchi, A.: Robustnerf: Ignoring distractors with robust losses. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 20626–20636 (2023)
20. Schonberger, J.L., Frahm, J.M.: Structure-from-motion revisited. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 4104–4113 (2016)
21. Turkulainen, M., Ren, X., Melekhov, I., Seiskari, O., Rahtu, E., Kannala, J.: Dn-splatter: Depth and normal priors for gaussian splatting and meshing. In: 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). pp. 2421–2431. IEEE (2025)
22. Umeyama, S.: Least-squares estimation of transformation parameters between two point patterns. IEEE Transactions on pattern analysis and machine intelligence **13**(4), 376–380 (1991)
23. Wang, Z., Bovik, A.C., Sheikh, H.R., Simoncelli, E.P.: Image quality assessment: from error visibility to structural similarity. IEEE transactions on image processing **13**(4), 600–612 (2004)
24. Wu, G., Yi, T., Fang, J., Xie, L., Zhang, X., Wei, W., Liu, W., Tian, Q., Wang, X.: 4d gaussian splatting for real-time dynamic scene rendering. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 20310–20320 (2024)
25. Yan, Y., Lin, H., Zhou, C., Wang, W., Sun, H., Zhan, K., Lang, X., Zhou, X., Peng, S.: Street gaussians: Modeling dynamic urban scenes with gaussian splatting. In: European Conference on Computer Vision. pp. 156–173. Springer (2024)
26. Yang, Z., Gao, X., Zhou, W., Jiao, S., Zhang, Y., Jin, X.: Deformable 3d gaussians for high-fidelity monocular dynamic scene reconstruction. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 20331–20341 (2024)
27. Ye, V., Li, R., Kerr, J., Turkulainen, M., Yi, B., Pan, Z., Seiskari, O., Ye, J., Hu, J., Tancik, M., et al.: gsplat: An open-source library for gaussian splatting. Journal of Machine Learning Research **26**(34), 1–17 (2025)
28. Zhang, R., Isola, P., Efros, A.A., Shechtman, E., Wang, O.: The unreasonable effectiveness of deep features as a perceptual metric. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 586–595 (2018)

29. Zhang, Z.: A flexible new technique for camera calibration. IEEE Transactions on pattern analysis and machine intelligence **22**(11), 1330–1334 (2000)
30. Zhao, C., Sun, S., Wang, R., Guo, Y., Wan, J.J., Huang, Z., Huang, X., Chen, Y.V., Ren, L.: Tclc-gs: Tightly coupled lidar-camera gaussian splatting for autonomous driving: Supplementary materials. In: European Conference on Computer Vision. pp. 91–106. Springer (2024)
31. Zhou, X., Lin, Z., Shan, X., Wang, Y., Sun, D., Yang, M.H.: Drivinggaussian: Composite gaussian splatting for surrounding dynamic autonomous driving scenes. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 21634–21643 (2024)

A Implementation details

Model and optimization. The canonical model uses $N{\approx}$18k Gaussians initialized from the SfM cloud after filtering (reprojection error $\leq 2\,$px, coordinate outliers removed). Colors are initialized by mapping the SfM point colors through an inverse sigmoid; scales are initialized near-isotropically to $\approx$1 cm; opacity logits are learnable. The deformation decoder is a 2-layer MLP (64 hidden units, ReLU) mapping $(\boldsymbol{\mu}_i, t)$ to the 10-dimensional offset $(\Delta\boldsymbol{\mu}, \Delta\mathbf{q}, \Delta\mathbf{s})$, evaluated in chunks. We optimize with Adam (learning rate 10^{-3} for Gaussian parameters, 10^{-4} for the decoder) for 15k iterations at 640×360; static-camera extrinsics are frozen (Sec. 3.1). Each iteration samples one camera uniformly and one time step; frames are decoded sequentially once and cached in memory. Table 4 summarizes all hyperparameters.

Table 4: Hyperparameters.

Static reference (Sec. 3.3)		**Training**	
voxel size v	5 cm	iterations	15k
frame stride k	2	render resolution	640×360
min. frames τ_f	15	Gaussians N	≈18k (no densif.)
sensor conf. threshold	$c \geq 2$	lr (Gaussians / MLP)	10^{-3} / 10^{-4}
GCS (Sec. 3.4)		**Loss (Sec. 3.5)**	
soft kernel σ	0.1 m	λ_d	0.1
hard gate τ	0.2 m	λ_g	0.1
photometric floor β	0.5	depth scale	mm → m

GCS precomputation. GCS maps are computed once per LiDAR frame at sensor resolution (256×192) using a k-d tree over the static reference, stored as a compressed archive, and nearest-upsampled to the render resolution during training. Frames whose depth or confidence maps are missing fall back to a neutral (all-ones) photometric weight, so training never interrupts. Building the static reference and all GCS maps takes a few minutes on CPU.

Evaluation details. CD is computed per evaluation frame between the deformed Gaussian centers and the back-projected LiDAR cloud of that frame; model points are cropped to the LiDAR cloud's bounding box (0.5 m margin) so that scene regions outside the single-view sensor frustum are not penalized, and the symmetric mean over both directions is averaged across frames. Static-region metrics use an off-the-shelf Mask R-CNN [5] person segmenter (score > 0.7, mask threshold 0.5, 3-pixel dilation); detected pixels are excluded for sPSNR, and copied from ground truth before computing sSSIM so that window statistics are unaffected at person boundaries. LPIPS uses the VGG backbone, matching the baseline protocol.

Runtime. A full training run completes in under one hour on a single consumer GPU (RTX 3090/4090 class); evaluation over all 1,275 held-out frames, including person segmentation, takes a comparable time.

The complete pipeline—alignment, static reference, GCS, training, and evaluation—runs end-to-end from a fresh checkout with three commands.